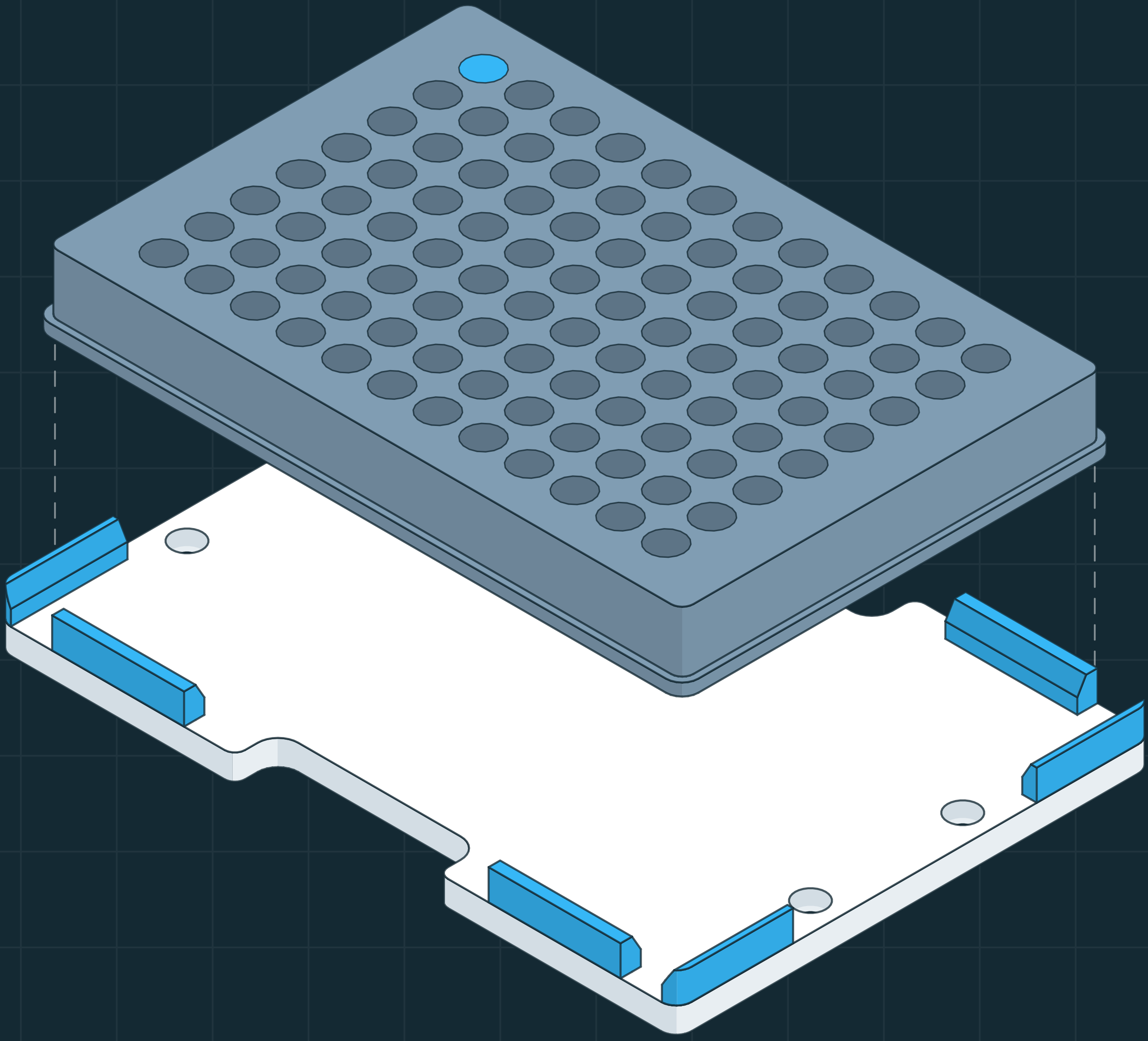
OpenNest

OPEN HARDWARE

# Design and manufacturing guide

V1.4 NOVEMBER 2026



OPEN HARDWARE / AGENT READY / ANSI/SLAS 1-2004

Adaptyv

● OPENNEST

# A nest for every SBS plate.

OpenNest is an open hardware deck plate that holds a standard microplate where a robot can find it. Tell an agent your hole pattern, then print or machine the file it builds.

| 134 × 94 | 128.3 × 86 | 2.0 | 30° |
|---|---|---|---|
| mm plate, 11 mm tall | mm pocket | mm catch per side | lead-in from vertical |

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## What has been tested

| V1.4 | STATUS |
|---|---|
| 3D print in PLA on a Bambu H2D | **Printed. No measurements recorded yet** |
| 3D print in other materials | Not tested |
| CNC aluminium | Not tested |
| CNC POM-C or PEEK | Not tested |
| CNC solid PTFE | Not tested |
| On a robot | Not tested |

● THAT IS THE WHOLE LIST

Every other number here is calculated from the model and the SLAS standard. Software checks read the file, not a real part.

● 01

# Every surface has a job.

A robot places a plate with some error. The nest absorbs that error, sets the plate down flat and leaves room for the gripper to get out.

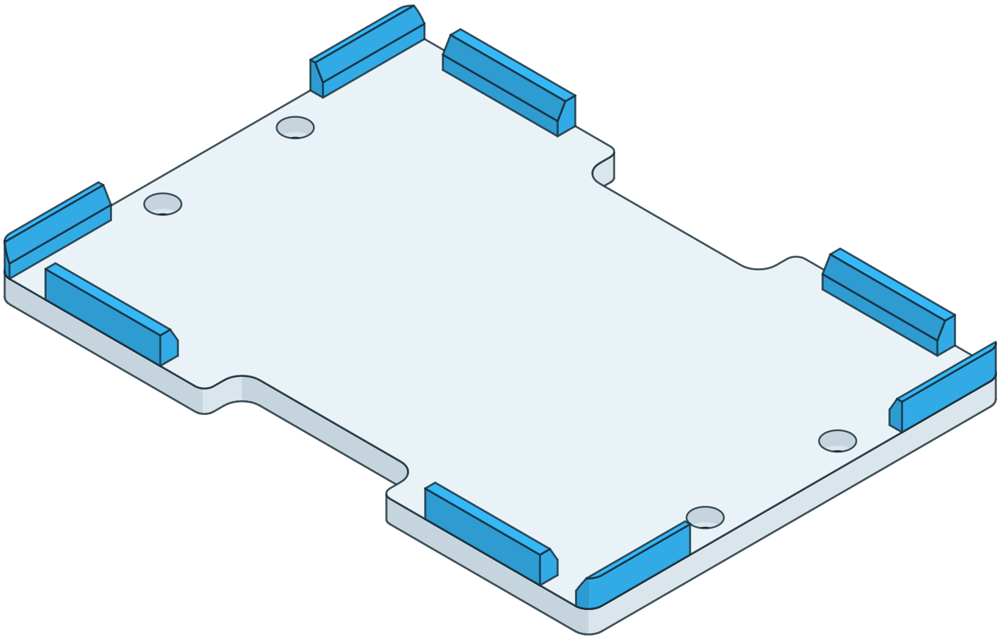


FIG 1 / THE PLATE, DRAWN TO SCALE FROM THE MODEL

## Catch

Eight tabs lead a plate in with a 30° slope. The slope starts 3 mm above the seat, so a nominal plate landing up to 2.0 mm off centre still finds the pocket.

## Seat

The walls meet the seat in a square corner. Nothing lifts the edge of the plate, so its height stays the height you taught.

## Release

Two 40 × 8.5 mm notches on the long sides give gripper fingers somewhere to go. The tabs stop short of them.

## Mount

Holes are yours to define. Screw heads sit below the seat, clear of the plate.

● BEFORE YOU RUN

Teach each nest on your own deck. Screw clearance lets a plate sit slightly differently from one position to the next, so do not copy positions between nests.

● 02

# Drawing and tolerances.

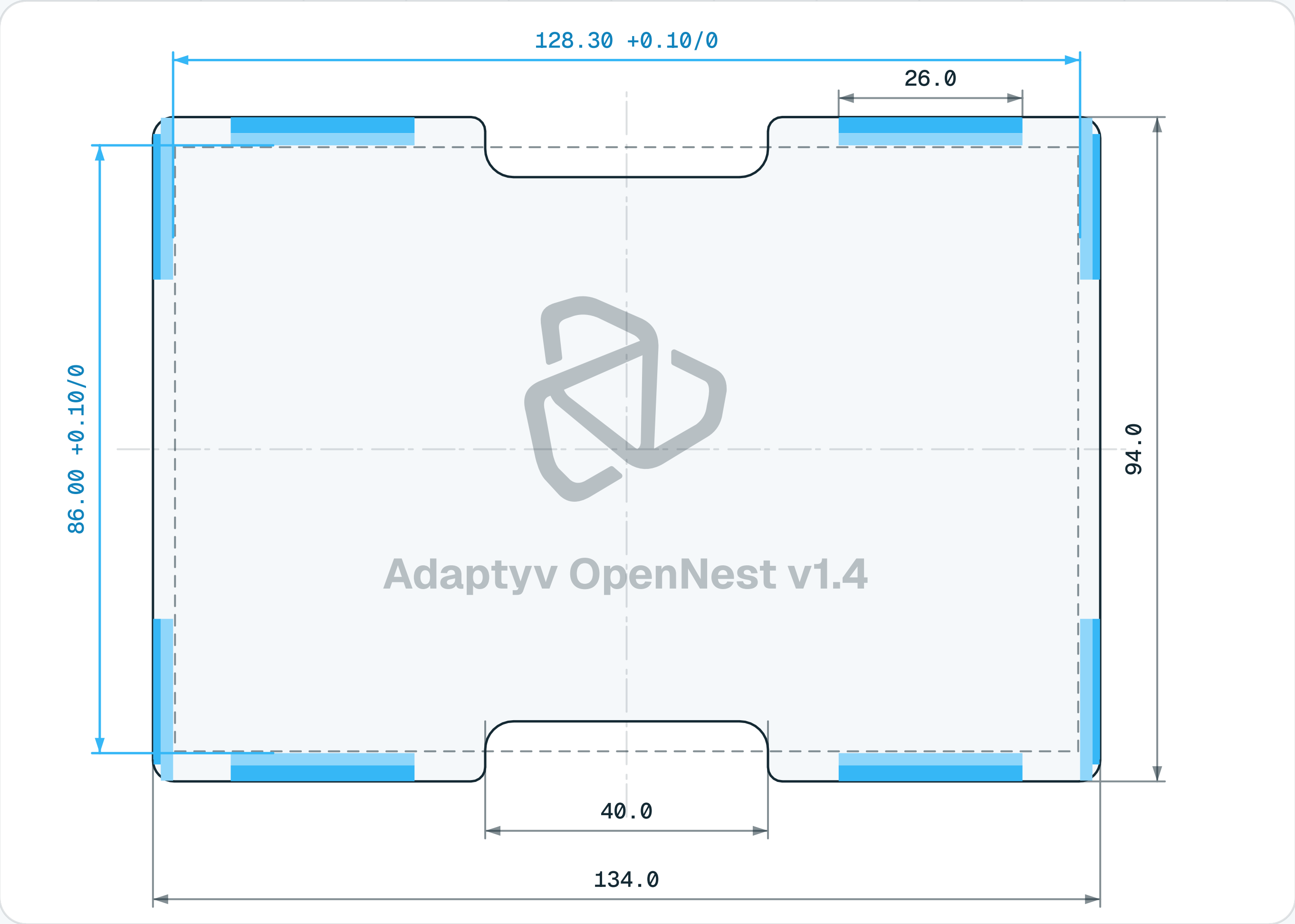

128.30 +0.10/0
26.0
86.00 +0.10/0
94.0
Adaptyv OpenNest v1.4
40.0
134.0

FIG 2 / PLAN VIEW. CYAN DIMENSIONS CONTROL THE FIT. DASHED LINE IS THE SLAS FOOTPRINT

| FEATURE | NOMINAL | TOLERANCE |
|---|---|---|
| Pocket, X | 128.30 | +0.10/0 |
| Pocket, Y | 86.00 | +0.10/0 |
| Seat height from underside | 5.00 | ±0.05 |
| Seat flatness | | 0.05 |
| Everything else | | ±0.1, ISO 2768-f |

● HOLES

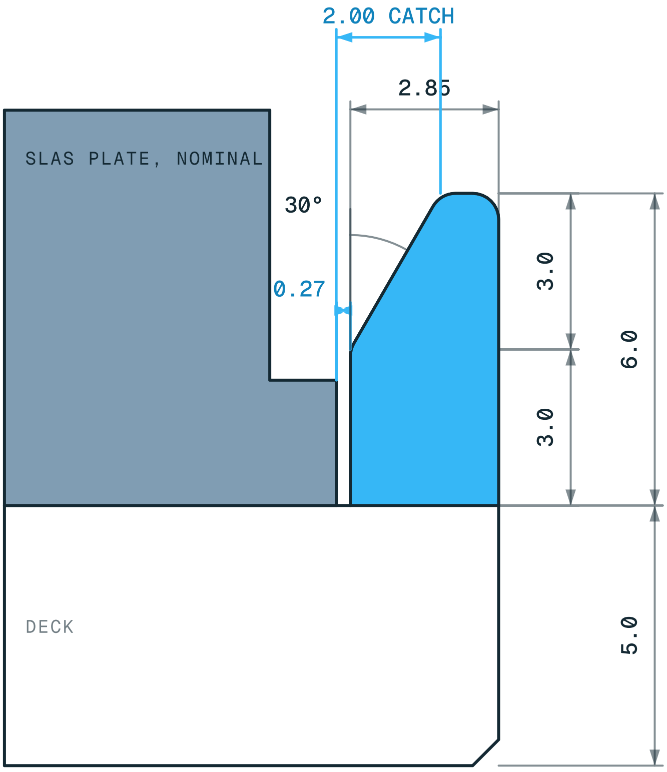
Hole positions are yours to define, so they are not on this drawing. The STEP file is the master.

03

# Fit against SLAS 1-2004.

By calculation, every plate the standard allows fits at 20 °C. Clearance is total, in mm.

| PLATE WITHIN SLAS | SIZE, X × Y | X CLEARANCE | Y CLEARANCE |
|---|---|---|---|
| Nominal | 127.76 × 85.48 | +0.54 | +0.52 |
| Largest within 12.7 mm of a corner, ±0.25 | 128.01 × 85.73 | +0.29 | +0.27 |
| Largest anywhere along a side, ±0.5 | 128.26 × 85.98 | +0.04 | +0.02 |



## 2.00 mm

Catch per side in X, 1.99 mm in Y

## ±0.27 mm

Plate float in X, ±0.26 mm in Y

## 30°

Lead-in from vertical

FIG 3 / SECTION THROUGH A CORNER TAB, NOMINAL PLATE AT 20 °C

LIMITS

Plastic plates grow faster than aluminium. By calculation, at 37 °C a plate at the ±0.5 mm limit is 0.06 mm too big in X and 0.05 mm in Y, and one within ±0.25 mm still clears by about 0.2 mm. Float of ±0.27 mm is comfortable for 96 and 384 wells. Check your tolerances before you run 1536-well plates.

● 04

# 3D printing.

A print proves the geometry, not the material. Measure it before you trust it.

| SETTING | VALUE |
|---|---|
| Process | 0.12 mm Fine, Bambu H2D, 0.4 mm nozzle |
| Lead-in, 7.88 to 11 mm above the bed | 0.08 mm layers |
| Walls and infill | 4 walls, 25% gyroid |
| Precise outer wall | On |
| Seam | Blocked on the tab faces a plate touches |
| Brim | Outer, 5 mm. Remove before measuring |
| Ironing | Off |
| Slicer estimate | About 2 h 45 min, about 50 g PLA |

## Check the print

01 Let it cool and remove the brim.

02 Measure the pocket between opposing tab walls, at the seat and just under the slope. Accept X 128.1 to 128.5 and Y 85.8 to 86.2.

03 If it is small, scale the object in XY by target / measured in the slicer and print again. A plate that only fails in an undersized printed pocket says nothing about a machined one.

04 Measure the seat height from the underside. The target is 5.0 mm. Check the seat is flat.

05 Then test what a print can prove: plates drop in along the lead-in and sit flat, holes line up with your deck, and the gripper clears.

● TESTED

v1.4 has been printed in PLA on a Bambu H2D. No other printer or material has been tried. PLA shrinks as it cools and softens near 60 °C, so use it for fit checks and room temperature work.

● 05

# CNC machining.

The STEP file is the part. A machine cuts it as drawn, with no shrink correction. No plate has been machined yet, so everything on this page is untested.

## Aluminium

| STEP | WHAT TO DO |
|---|---|
| Stock | 12 mm rolled plate, EN AW-6082-T651 or 6061-T651. Not 5083 cast plate, which anodises grey and blotchy. |
| Op 1, top | Clear the seat around the tabs. Finish the tab walls with a square end mill. Cut the 30° lead-in, drill your holes, then profile the outline and notches. |
| Op 2, flip | Break the underside edge 0.5 mm and remove the holding tabs. |
| Finish | Anodize per MIL-PRF-8625, Type II, Class 2. Dimensions apply after coating. |
| Engraving | 0.3 mm deep, engraved before or after coating. Say which on the order. |

● HARDCOAT

R0.5 edges suit Type II. Thick Type III needs larger edge radii, so change the model before you order it.

## Other materials

| MATERIAL | POCKET | WHAT TO KNOW |
|---|---|---|
| POM-C, PEEK | POM-C 128.4 × 86.1.<br>PEEK 128.3 × 86.0 | Polymers creep under screw heads. Keep torque low and recheck it. |
| PTFE-impregnated anodise on aluminium | As aluminium | Low friction surface that keeps aluminium's stability. Check the edge radii first. |
| Solid PTFE | 129.0 × 86.5 | Changes about 0.4 to 0.6 mm over the pocket between 19 and 23 °C. The script sizes a wider pocket and bores for metal sleeves. A nominal plate floats about ±0.6 mm, so use it for 96-well plates only. |

● 06

# Agent ready.

The pocket, tabs and notches stay fixed. An LLM agent reads the spec, runs the script with your hole pattern and writes your files.

EXAMPLE PROMPT

```
Make me an OpenNest with four M3 cap head screws,
110 × 60 mm apart. I'm printing it in PLA on a Bambu H2D.
```

WHAT THE AGENT RUNS

```
uv run cad/nest.py make --material pla --screw M3 --head cap --spacing 110x60 \
  --name deck-a --label "Slot A3"
```

## Tell the agent

| ITEM | WHAT TO GIVE IT |
|---|---|
| Holes | Positions from the plate centre, or the spacing between them. |
| Screws | Size and head: cap head or flat head. |
| Process | 3D print, CNC aluminium or another CNC material. |
| Material | What you will make it from, so it can size the pocket. |
| Label | Optional free text engraved under the title and version, such as a slot name. |

## How it runs

01 Open the repo in your coding agent.

02 Describe your holes and your process.

03 The agent runs the script, then checks that the holes clear the tabs and the plate.

04 Measure your deck before you cut. The script checks holes against the plate, not against your hardware, and it reads the file, not a real part.

● KEEP THE RECORD

Every build writes nest.toml: the version, your inputs and every dimension. Keep it with the part. To build the same part again, run make --config nest.toml.

● 07

# Acceptance check.

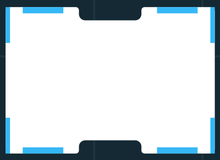
Fill this in for each part before it goes on a deck.

| | | | |
|---|---|---|---|
| PART | MATERIAL | MEASURED BY | DATE |
| | | | |

| CHECK | TARGET | MEASURED |
|---|---|---|
| Pocket X, at the seat | 128.30 +0.10/0 | |
| Pocket Y, at the seat | 86.00 +0.10/0 | |
| Pocket X, printed part | 128.1 to 128.5 | |
| Pocket Y, printed part | 85.8 to 86.2 | |
| Seat height from underside | 5.00 ±0.05 | |
| Seat flatness | 0.05 | |
| Envelope | 134 × 94 × 11 | |
| Plate drops in by hand without sticking | Pass | |
| Plate sits flat and does not rock | Pass | |
| Screw heads sit below the seat | Pass | |
| Holes line up with your deck | Pass | |
| Robot places and picks without contact | Pass | |

● SHARE IT

Open an issue with your numbers, printer or shop, and material. Real measurements are what this design needs next.



# Build it. Measure it. Share what you find.

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OPEN HARDWARE / AGENT READY

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